TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Maximilian König	**Email**	maximilian.koenig@igcv.fraunhofer.de
Institute Name	Fraunhofer Institute for Casting, Composite and Processing Technology IGCV			
Address	Am Technologiezentrum 10, 86159 Augsburg, Germany			

2. Technology Overview

Title of Proposed Technology	Augmented Reality–Based Teach-In and Skill Transfer for Industrial and Service Robots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Mechanical・Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	Software	Embedded Software
Concept of Proposed Technology	The proposed technology develops an Augmented Reality (AR)–based teach-in and skill transfer framework that enables intuitive programming as well as data set generation of industrial and collaborative robots in high-mix, low-volume assembly environments. The core concept is to allow human workers to teach robot tasks directly in the real workspace through guided demonstration, eliminating the need for expert-level robot programming. The system's main functions include AR-based task guidance, capture of human motions and spatial constraints, transformation into structured robot task representations, and real-time visualization of robot states during validation and execution. The approach focuses on constraint-aware task abstraction and the reuse of taught skills across product variants, enabling robust task adaptation without low-level robot reprogramming. Beyond direct teach-in, the framework also supports AR-based teleoperation of robots, allowing			

	users to intuitively guide robot motions in real time. This interaction enables the systematic capture of human demonstrations and robot trajectories, which can be used to generate high-quality datasets for data-driven and AI-based robot learning approaches, such as learning-from-demonstration or teleoperation-assisted training.
Rationale for Joint R&D Project	High-mix, low-volume production and frequent product changes make robotic automation economically challenging for many SMEs. The main barrier is not robot hardware, but the effort and expertise required for task programming, re-programming, calibration, and validation whenever products, fixtures, or process parameters change. This leads to long commissioning times and a high dependence on specialists, limiting the adoption of robotic solutions in both industrial and non-industrial application domains. On the other hand, AI solutions for highly flexible automation (e.g., VLA) remain limited due to the amount of task-specific real data required. Augmented reality-based robot programming offers solutions for both intuitive teach-in and data set generation for AI applications. A joint R&D approach is required because the key capabilities needed to develop a research-oriented and industrially relevant solution are complementary. Fraunhofer IGCV, as the research institute, contributes expertise in AR interaction design, human-in-the-loop teach-in methods, task abstraction, software architecture, and systematic evaluation. Building on prior work, Fraunhofer IGCV has already developed and evaluated adaptive AR-based qualification and assistance systems that enable competence-oriented, stress-reduced learning and data-driven analysis of learning curves. This validated experience provides a strong methodological foundation for the proposed approach, particularly with respect to human-centred interaction, adaptive assistance, and iterative system improvement. In addition, Fraunhofer IGCV combines long-standing expertise in augmented reality–based worker assistance with robotics and automation research, including the modelling of assembly tasks, process constraints, and human actions in structured and machine-interpretable representations. This expertise extends to automated robot programming and learning approaches, such as imitation learning and reinforcement learning, where human demonstrations serve as a key input for skill acquisition. These capabilities are directly transferable to AR-based robot teach-in, teleoperation, and skill transfer, enabling the generation of structured datasets that can be used for data-driven and AI-based robot learning, including vision–language–action (VLA) models. The Korean SME partner represents research-oriented SMEs or start-ups that actively develop their own robotic technologies and software solutions rather than acting solely as system integrators. These partners contribute access to SME-relevant robot platforms, controller interfaces, and application scenarios, as well as domain-specific development requirements. Typical partner profiles include companies working on intuitive robot programming, AI-based robot learning using data sets via teleoperation, or system-level development of industrial and service robotic solutions. Through iterative co-development and joint validation, the framework can be advanced beyond a

	conceptual level toward a functionally integrated TRL 5 prototype that supports the partners' own technology development and future commercialization activities.
Key Words	1. Augmented Reality 2. Robot Teach-In 3. Skill Transfer 4. Human-Robot Interaction 5. High-Mix Assembly

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The objective of this joint R&D project is to develop and validate an AR-based teach-in and skill transfer framework that significantly reduces the effort and expertise required to program and adapt industrial and service robots in complex, variant-rich work environments. The project aims to achieve the following specific objectives: ▪ Develop a human-centred AR teach-in workflow that enables non-expert users to demonstrate robot tasks directly in the real workspace. ▪ Design and implement a software framework that captures human actions, spatial relations, and process constraints and translates them into structured robot task representations. ▪ Integrate the AR-based teach-in approach with robot controllers and end-effectors relevant to industrial and service robotic systems, ensuring compatibility with SME-oriented development platforms. ▪ Enable bidirectional feedback through AR visualization for task validation, execution monitoring, and rapid adaptation to changing tasks and environments. ▪ Validate the developed prototype in realistic, SME-oriented application scenarios, demonstrating reduced programming effort, increased flexibility, and improved usability. Together, these objectives bridge human-centred AR interaction with automation-oriented task abstraction, establishing the basis for robust robot teach-in and skill reuse across industrial and service robotics domains, and advancing the technology from laboratory validation toward a functionally integrated prototype (TRL 5) suitable for further industrial deployment and commercialization.			
R&D Target Specifications	The R&D project targets the following measurable performance specifications for the AR-based teach-in and skill transfer framework: ▪ Significant reduction of robot programming and reconfiguration time compared to conventional teach pendant or offline programming approaches, targeting a reduction in the range of 40–60 %. ▪ Enable task teach-in and adaptation by non-expert users, demonstrating a clear decrease in required programming expertise and training effort. ▪ Support flexible adaptation to product variants and process changes without modification of low-level robot code. ▪ Achieve stable and repeatable task execution through integration with industrial robot controllers and end-effectors under realistic assembly conditions. ▪ Provide interactive AR-based feedback that supports teach-in, task validation, and execution monitoring in real production environments. ▪ Provide an interactive AR-based method for teleoperation to generate real robot data for AI-training The target specifications will be evaluated using representative SME-oriented assembly scenarios and measurable indicators such as programming time, number of required iterations, task completion consistency, and structured user feedback.			
R&D Plan	The R&D project follows a structured, human-centred research and development approach to advance the proposed technology from laboratory validation toward a functionally integrated prototype suitable for SME-oriented assembly environments. The development process is aligned with the principles of DIN EN ISO 9241-210, ensuring systematic consideration of user needs, task contexts, and iterative evaluation.			

	Phase 1: System Concept and AR-Based Teach-In Design Analysis of users, tasks, and operating environments; definition of system architecture; design of AR-based teach-in workflows and interaction concepts. Methodology: Human-centred requirements analysis and concept design. Milestone: Defined system architecture and validated AR teach-in concept. **Phase 2: Software Framework Development and Robot Integration** Implementation of a modular software framework; capture and abstraction of human actions, spatial relations, and process constraints; integration with industrial robot controllers and end-effectors. Methodology: Iterative prototyping and system integration. Milestone: Functionally integrated AR-based teach-in prototype. **Phase 3: Prototype Validation and Evaluation** Validation of the prototype in realistic, SME-oriented assembly scenarios; assessment of programming effort, flexibility, usability, and robustness. Methodology: Iterative evaluation in line with human-centred design principles. Milestone: Validated prototype demonstrating readiness for further industrial development. The project timeline follows this phased structure, ensuring a systematic progression toward a functionally integrated prototype at TRL 5 within the project duration.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IGCV is responsible for the conceptual design, methodological development, and technical realization of the AR-based teach-in and skill transfer framework. This includes: ▪ Design of the human-centred AR interaction concepts and teach-in workflows in line with established usability and human-centred design principles. ▪ Development of the core software framework, including task abstraction, skill representation, and AR-based feedback mechanisms. ▪ Definition of evaluation methods and performance indicators to assess usability, flexibility, and robustness. ▪ Technical coordination of the joint R&D activities and consolidation of results toward an integrated prototype. Through these contributions, the research institute ensures the scientific soundness, methodological rigor, and technological innovation of the proposed solution.
Expected R&R of Korean SME Partner	The Korean SME partner is responsible for contributing application-driven research input, technology development perspectives, and validation support for the proposed AR-based teach-in framework. Rather than acting solely as a system integrator, the partner represents a research- and development-oriented SME or start-up that actively develops its own robotic software, control concepts, or AI-based solutions. This includes: ▪ Provision of industrial and/or service robot platforms, controller interfaces, and development environments relevant to SME-oriented R&D activities. ▪ Contribution to the development and evaluation of novel robot programming, teleoperation, or learning-based approaches, including the use of AR-based demonstrations as training data for AI methods. ▪ Definition of representative use cases and experimental scenarios that reflect both industrial and non-industrial application domains, such as manufacturing, service robotics, or assistance systems. ▪ Participation in iterative prototype testing, evaluation, and refinement cycles to assess usability, flexibility, and suitability for further technology development and commercialization. Through these contributions, the Korean SME partner ensures that the developed solution addresses real research and development challenges faced by innovative SMEs, supports the creation of proprietary technologies, and provides a foundation for future product and platform development. Both partners jointly contribute to aligning the technical development with SME-oriented research requirements, iteratively refining the prototype based on evaluation results, and preparing the outcomes for further technology development, knowledge transfer, and commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this joint R&D project can be summarized in terms of technological achievements, commercialization potential, and broader industrial and economic impact. **Technological achievements:** The project delivers a functionally integrated AR-based teach-in and skill transfer prototype that enables intuitive programming and adaptation of industrial and service robots in complex and dynamic work environments. The developed framework improves flexibility, reduces commissioning effort, and enhances usability, thereby advancing the state of the art in human-centred robotic automation. The figure below illustrates the envisioned application scenario. It shows a researcher from the Fraunhofer Institute for Casting, Composite and Processing Technology (IGCV) interacting with a robotic system using gesture-based control in an augmented reality environment, exemplifying the concept of intuitive, human-centred robot teach-in and control. **Commercialization potential:** The validated prototype provides a strong basis for further technology development, customization, and productization by Korean SMEs and start-ups. The modular and robot-agnostic design enables integration into existing automation and robotics ecosystems, supporting the development of market-ready solutions and services tailored to SME needs. A key exploitation pathway is the transfer of developed concepts, methods, and software components through licensing models, enabling Korean SMEs to integrate the AR-based teach-in framework into their own robotic products, platforms, or AI-driven solutions while building proprietary extensions. Beyond commercialization, the joint R&D project strengthens strategic collaboration between Germany and Korea in key technology domains such as robotics, artificial intelligence, and human-centred automation. By combining complementary research and development strengths, the collaboration contributes to technological sovereignty and long-term innovation capacity of both countries' SME ecosystems, enabling the development of independent, high-value technologies in an increasingly competitive global landscape.

	Broader industrial and economic impact: By lowering the barrier to robotic automation, the project supports productivity improvements, faster response to product variants, and more efficient use of automation technologies in SME manufacturing. In addition, the joint R&D approach strengthens international collaboration and technology transfer, contributing to sustainable innovation and long-term value creation within SME-oriented industrial ecosystems.

6. References

(1) DIN EN ISO 9241-210 – Ergonomics of human-system interaction – Human-centred design for interactive systems.

(2) König, M., Stadlmaier, M., Rusch, T., Sochor, R., Merkel, L., Braunreuther, S., & Schilp, J. (2019).
MA²RA – Manual Assembly Augmented Reality Assistant.
In: Proceedings of the 2019 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM). Macao, China: IEEE.
https://doi.org/10.1109/IEEM44572.2019.8978844

(3) König, M., Merkel, L., Arndt, B., Berger, C., Heller, M., Eisenbarth, M., & Schilp, J. (2023).
Development of an Adaptive Augmented Reality Qualification System for Manual Assembly and Maintenance.
In: Proceedings of the 5th Conference on Production Systems and Logistics (CPSL 2023).
https://doi.org/10.15488/15312

(4) König, M., Neumeister, B., Arndt, B., Härdtlein, C., Merkel, L., Teubert, A., Halbauer, M., Heller, M., Eisenbarth, M., & Schilp, J. (2024).
Evaluation of an Augmented Reality Qualification System for Manual Assembly and Maintenance.
In: Proceedings of the 6th Conference on Production Systems and Logistics (CPSL 2024).
https://doi.org/10.15488/17754

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization